

the devices have to have a clear line of sight between the transmitting and receiving stations in order to work. These laser systems operate in the terahertz (THz) frequency spectrum and usually operate in the 194 THz or 375 THz range. Their performance is comparable to the best fibre optic system available, giving speeds between 622 Mbps and 1.25 Gbps. Research predicts that this can go as high as 160 Gbps—faster than your hard disk's interface!

Another advantage of this technology is that it's portable and fast to implement. Conventional high-speed connections like T1 lines take between two to four months to install as the roads have to be dug up to lay down the cables, not to mention the bureaucratic formalities involved. On the other hand, FSO only involves mounting the transmitting and receiving units between buildings. FSO devices look a lot like security video cameras and can be mounted on rooftops, corners of buildings or even inside offices behind windows. Another inherent advantage with this system is that of security—it is exceedingly difficult to jam or intercept an optical signal without breaking it, thus alerting the user.

However, FSO does have its disadvantages. Interference with the line of sight is one of the biggest drawbacks. If the laser beam is broken by a passing bird or heavy rain, it will affect the bandwidth or even halt data transfer completely. Even extreme weather conditions like fog and mist diffuse the beam and disrupt data transmission. In high-rise constructions, the wind causes the buildings to sway, affecting the alignment of the laser beam if the equipment is mounted on top of the structure. Service providers have created multiple workarounds for these problems that basically work at maintaining the alignment of the transmitter and receiver in real time. Such systems have been used successfully in ships that were able to maintain connectivity with hubs located on the shore. However, these work out to be much more expensive than conventional FSO systems.

As with any other budding technology, in order to get it out of the laboratories and into the realm of mass acceptance, it has to establish its reliability. Most FSO companies demonstrate a reliability

of 99.99 per cent at distances of up to 2 miles, compared to the magical 99.999 per cent expected of broadband connectivity. It is this 'five nines' reliability that still eludes most implementers of this technology.

As FSO evolves, its ease of implementation and cost-friendliness breeds the possibility of connecting rural areas, where laying fibre optic cables is economically unadvisable. It can also be applied to cellular phone technology, paving the way for broadband 3G cellular networks or even as a temporary network for disaster management (*see box, 'FSO to the Rescue'*).

The power within

Every few years, an innovative use of an existing technology bridges the gap between the old and the new, making us think, "Gee, why didn't anyone think of this before?" Power

Line Communications (PLC) is one such technology. Using PLC, various companies have used ordinary power lines such as those found in homes and offices to transmit data.

Theoretically, the maximum bandwidth achievable with PLC is 14 Mbps, and the practical speeds range between 6 to 8 Mbps, making it slightly faster than WiFi (Wireless Fidelity) networks. WiFi is a standard for devices compliant with the 802.11b wireless standard (*for more information, see 'Wireless Connectivity' in Digit, June 2002*).

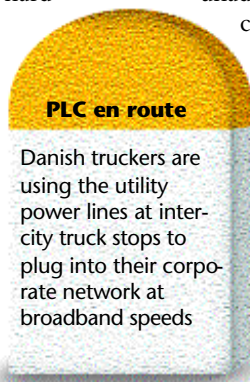
But once networking enters the realm of utility power lines, it is in very harsh territory, electrically. It will have to address the issues of static, noise and high

voltage before it is looked upon as a viable networking system. The evolution of Application Specific Integrated Circuits (ASICs) is the main driver of PLC technology. ASICs are basically small processors with powerful signal processing capabilities, which filter out the noise in the power lines. Unlike a traditional Ethernet-based network using CAT5 cable that is far more controlled in terms of shielding, power, voltage levels and impedances, utility power lines present a nightmare of situations that need to be dealt with for reliable data transfer.

One of the biggest hurdles in home PLC is the varying transfer function of the power circuitry. The transfer function is the overall response of the power lines, taking into account the different impedances and the levels of voltage all over the lines—the transfer function will vary if you plug in a new appliance or switch power supplies. To counter these situations, a technique that allows adaptive response to variable conditions in the power line was developed. HomePlug is one such standard that allows home networking through power lines. It employs an adaptive approach to get the highest possible data transfer rate on the channel by varying the bit rate of the information in accordance with the noise levels and the condition of the power line.

Another concern is that of security—since most homes in a neighbourhood are connected through a single distribution grid, there exists a distinct possibility of your neighbour being able to pry into your network. Some manufacturers have integrated passwords into their home PC gear to solve this problem.

PLC removes the need for an additional network between your home and



FSO to the Rescue

One of the most critical implementations of Free Space Optics (FSO) technology was seen during the September 11 tragedy, where various FSO services provided the much-needed bandwidth to the rescue teams. Its ease of installation made FSO the only viable technology to respond so quickly to the bandwidth requirements. FSO companies such as LightPointe and Terabeam restored gigabit connectivity within a week to companies such as Merrill Lynch, which was losing millions of dollars each disconnected day and Federal



WTC Twin Towers

Emergency Management Agency, which carried out the rescue operations.